

Software Engineering 11–12 | Glossary of Terms

A

algorithm	A step-by-step procedure required to solve a problem. Algorithms may be presented in many ways, e.g. written instructions, flow charts or using a computer programming language.
app	A software program designed for a specific purpose to run on mobile devices or on a personal computer. An abbreviation of the word 'application'.
application programming interface (API)	An interface that allows an application or website to plug into another program or website.
artificial intelligence (AI)	Intelligence demonstrated by machines. Sometimes called machine intelligence.
assets	The digital elements that can be used or incorporated into a multimedia presentation or production. They include graphics, photographs, videos, audio, animations or other artistic data.

B

big data	Extremely large datasets that may be analysed computationally to reveal patterns, trends and associations.
black box testing	A testing technique where there is no knowledge of the internal workings of an application. A user will provide inputs and observe the outputs in order to determine functionality of the program.

C

collaborate	Working with others towards a shared goal, through a variety of modes of communication. This may be achieved using a range of technologies, tools and processes.
computational thinking	A process in which a problem is analysed and solved so that a human, machine or computer can effectively implement the solution. It involves using strategies to organise data logically, break down problems into parts, interpret patterns and design, and implement algorithms to solve problems.

D

data	A discrete representation of information using number codes. Data may include characters (e.g. alphabetic letters, numbers and symbols), images, sounds and/or instructions that, when represented by number codes, can be manipulated, stored and communicated by digital systems.
data sovereignty	A global movement that refers to the right of Indigenous Peoples to determine the creation, collection, ownership and application of data that is for and about Indigenous Peoples. The term also refers to transnational data flow and a range of issues arising from the flow of data between nations.

data types	Data types used in computing are expressed as either: string/text, character, integer, floating point or real, date and time, Boolean.
design thinking	A process where a need or opportunity is identified and a design solution is developed. The consideration of economic, environmental and social impacts that result from designed solutions are core to design thinking.
digital technologies	Refers to electronic tools, systems, devices and resources that generate, process or store data, and may include applications, games, microcontrollers, mobile devices, multimedia, networks, robotics.

E

evaluate	Make a judgement based on criteria; determine the value of.
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G

grey box testing	- A testing technique where there is a partial knowledge of the internal workings of an application. Its purpose is to search for defects related to improper code structure or the improper use of functions. This technique gives the ability to test both the code and the presentation layer of an application. - Note: grey box testing and gray box testing mean the same.
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I

information	The presentation of data in a manner that is readily understood.
information system	The combination of digital systems, people and processes that collect, manage and analyse data.

M

model	A mathematical, conceptual or physical representation that describes, simplifies, clarifies or provides an explanation of the structure, workings or relationships within an object, system or idea. Models can provide a means of testing and predicting behaviour within limited conditions.
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O

object-oriented programming (OOP)	A paradigm based on the concept of 'objects' that can contain data and code in the form of procedures. OOP language is a language based on the principles of 'C'.
Object-Relational Mapping (ORM)	A technique used in object-oriented programming to query and manipulate data from a database.

P

procedural programming	A method of programming where the program is divided into functions. A program consists of data and procedures (modules) that operate on the data. Data and procedures are treated as separate entities.
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prototype	A trial product or model built to test an idea or process to inform further design development. Its purpose is to see if and how well the design works and is tested by users and systems analysts. A prototype can be either a physical object or exist in digital form.
pseudocode	A form of algorithm description that uses English-like statements with defined rules of structure and keywords.

R

refinement charts	A diagrammatic model that breaks down the steps of an outline solution into smaller and smaller steps.
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S

scripting language	A programming language designed for integrating and communicating with other programming languages.
structured data	Data with a high degree of organisation, such that it can easily be analysed and processed using algorithms. The organisation of the data takes on a form that captures the properties and behaviours of the source the data represents.
structured query language (SQL)	Specialist programming language used to manage data and access data in relational database management systems.
systems thinking	An understanding of how related objects or components interact to influence how systems function. Understanding the complexity of systems and the interdependence of components is important for scientific research and for the creation of solutions to technical, economic and social issues.

U

user experience (UX)	The usability, ease of use, and enjoyment provided in the interaction between the customer and the product.
user interface (UI)	The means by which users interact with computer hardware or software. In software, this usually comprises fields for text and number entry, mouse pointers, buttons and other graphical elements.

W

W3C	The World Wide Web Consortium is the international standards organisation for the world wide web. All of the standards used for web development are defined by this organisation.
white box testing	A testing technique used to examine the internal workings of an application including the design, code and inner workings of software. This technique gives the ability to verify the flow of inputs and outputs through the application.